LESSON 6.9

Divide Whole Numbers by Fractions

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing a whole number by a fraction means — how many pieces fit, or how much is in one whole — find the quotient with a model or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Park has 4 pounds of evidence powder. Each testing bag holds exactly 1/2 pound. He needs to prepare as many testing bags as possible before the lab deadline. How many bags can he fill?

REMEMBER

Divide Whole Numbers by Fractions

Dividing Whole Numbers by Fractions. Formula: a ÷ (b/c) = a × (c/b) Dividing a whole number by a fraction asks how many of those pieces fit, or how much is in one whole. 4 ÷ 1/2 counts halves: every whole holds 2 halves, so the answer is 4 × 2 = 8. That is why the shortcut called Keep, Change, Flip works — dividing by a fraction gives the same answer as multiplying by its reciprocal. Check by multiplying: 8 × 1/2 = 4.

1. Name the question: 4 ÷ 1/2 asks how many 1/2s fit in
2. Count with a model: each whole holds 2 halves, so 4 wholes hold 4 × 2 = 8 halves.
3. Shortcut: multiply by the reciprocal of the divisor, 6 ÷ 3/4 = 6 × 4/3 = 8, then check by multiplying: 8 × 3/4 = 6

Words: Whole Number *(Número entero)* · Fraction *(Fracción)* · Reciprocal *(Recíproco)* · Keep, Change, Flip *(Mantener, cambiar, invertir)* · Quotient *(Cociente)*

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which expression is equivalent to n × 1?

(A) 1 (B) n + 1

(C) n (D) 0

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A 5-foot strip of evidence tape is 1/3 of a full roll. How long is the full roll? (5 ÷ 1/3 asks how much is in one whole roll.)

(A) 5/3 feet

(B) 8 feet

(C) 1/15 foot

(D) 15 feet

HINT The 5 feet is only one third of the roll.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which property is shown? 7 × 5 = 5 × 7

(A) Associative Property

(B) Identity Property

(C) Commutative Property

(D) Distributive Property

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which property is shown? (2 + 9) + 1 = 2 + (9 + 1)

(A) Associative Property

(B) Commutative Property

(C) Identity Property

(D) Distributive Property

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which grouping makes 4 × 25 × 7 easiest to multiply mentally?

(A) (4 × 25) × 7

(B) 4 × (25 + 7)

(C) (4 + 25) × 7

(D) (25 × 7) × 4

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A detective has 10 feet of rope to section off a crime scene. Each section needs to be 1/4 of a foot long for detailed evidence markers. How many markers can the detective set up? 10 ÷ 1/4 = ?

(A) 2.5

(B) 4

(C) 14

(D) 40

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does 3 ÷ 1/4 = 12? How does the number line show this?

SENTENCE STARTER 3 ÷ 1/4 = 12 because it takes ___ jumps of ___ to get from 0 to ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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WHY THIS MATTERS Agent Park has 4 pounds of evidence powder. Each testing bag holds exactly 1/2 pound. He needs to prepare as many testing bags as possible before the lab deadline. How many bags can he fill?

1 **Circle the letter of the best answer. Show how you know.**

A detective has 10 feet of rope to section off a crime scene. Each section needs to be 1/4 of a foot long for detailed evidence markers. How many markers can the detective set up? What is the reciprocal of 1/4?

(A) 1/4
(B) −1/4
(C) 1/2
(D) 4

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A detective has 10 feet of rope to section off a crime scene. Each section needs to be 1/4 of a foot long for detailed evidence markers. How many markers can the detective set up? A student answered 2.5. What did they do wrong?

(A) They multiplied by 4
(B) They added 1/4
(C) They used the wrong rope length
(D) They divided by 4 instead of by 1/4

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Detective Park has 4 cups of flour. Each batch of evidence-cookies needs 1/2 cup. How many batches can he make?

(A) 2
(B) 4
(C) 6
(D) 8

Rewrite each fraction _____

Work it out _____

Simplify _____

4 **Circle the letter of the best answer. Show how you know.**

The crew has 8 quarts of sports drink. Each detective's water bottle holds 4/5 of a quart. How many bottles can be filled?

(A) 5
(B) 10
(C) 12
(D) 32/5

Rewrite each fraction _____

Work it out _____

Simplify _____

5 **Circle the letter of the best answer. Show how you know.**

There are 3 bags of trail mix. Each snack pouch needs 2/3 of a bag. How many full pouches can be filled, and is there a leftover?

(A) 2 pouches
(B) 4 1/2 pouches (4 full, half a pouch left)
(C) 4 pouches exactly
(D) 6 pouches

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Detective Park bakes with 9 cups of batter. Each muffin pan holds 3/4 of a cup. How many pans can he fill?

(A) 6
(B) 9
(C) 12
(D) 27/4

Rewrite each fraction _____

Work it out _____

Simplify _____

EXPLAIN YOUR THINKING

Why does 3 ÷ 1/4 = 12? How does the number line show this?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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